

2016.0 RANGE ROVER (LG), 501-28

ROOF SHEET METAL REPAIRS

ROOF PANEL (G1454610)

REMOVAL AND INSTALLATION

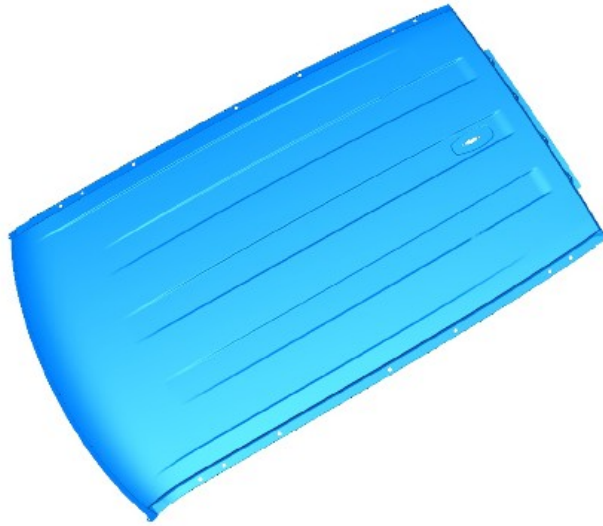
REMOVAL

NOTE:

In this procedure the roof panel is installed in conjunction with:

- Windshield glass
 - Liftgate
 - Headliner
 - Roof moulding
 - Side air curtain module
 - Antenna
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1.



E143050

The roof panel is serviced as indicated.

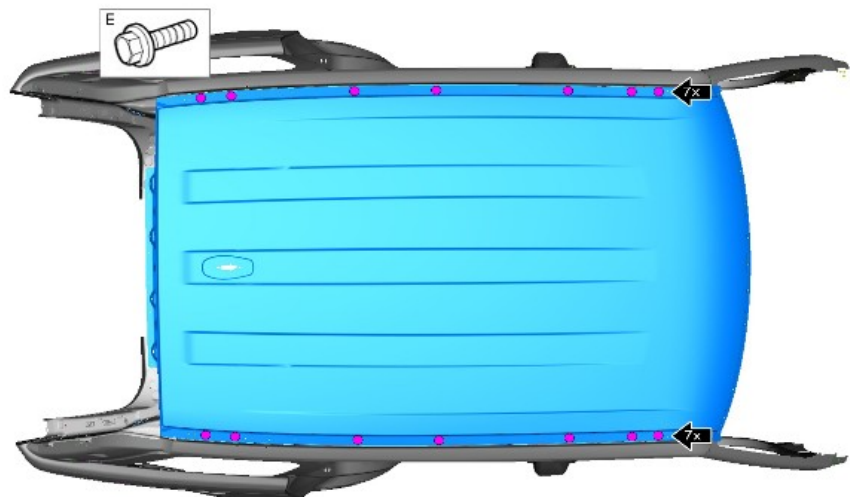
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2. Before commencing this procedure make sure that you are aware of all Health and Safety requirements. For additional information, refer to: [Health and Safety Precautions](#) (100-00 General Information, Description and Operation).
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3. For further information on the methods, tools and fixings used in this procedure refer to the body repairs - general information section.

NOTES:

- Where breakstem fasteners (BSF) removal is instructed remove fastener shaft with a 4mm punch and remove the head with a 6.5mm drill bit. Make sure all the drilled holes are free from debris.
- Where self piercing rivets (SPR) removal is instructed use either the approved SPR tool or 5.3mm drill bit.
- Any joints sealed as part of vehicle manufacture must be sealed as part of the repair.

4. For additional information, refer to: [Windshield Glass](#) (501-11 Glass, Frames and Mechanisms, Removal and Installation).
5. For additional information, refer to: [Liftgate](#) (501-03 Body Closures, Removal and Installation).
6. For additional information, refer to: [Headliner](#) (501-05 Interior Trim and Ornamentation, Removal and Installation).
7. For additional information, refer to: [Side Air Curtain Module](#) (501-20B Supplemental Restraint System, Removal and Installation).
8. For additional information, refer to: [Roof Moulding](#) (501-08 Exterior Trim and Ornamentation, Removal and Installation).
9. For additional information, refer to: [Audio Unit Antenna](#) (415-01 Information and Entertainment System, Removal and Installation).

10.



E 144832

Remove the bolts as indicated.

11. Remove any electrical components in the local area of repair to prevent damage.

12. Release and position the roof panel wiring harness to one side.

13.



E143051

Remove the SPR's as indicated.

14. Separate the joints and remove the old panel.

INSTALLATION

CAUTION:

Where structural adhesive is to be applied between surfaces it is essential compression is applied to the joints during the curing period.

NOTES:

- BSF installation involves using the Land Rover approved BSF installation tool and requires a hole diameter of 6.5mm.
- SPR installation requires using the Land Rover approved SPR tool.

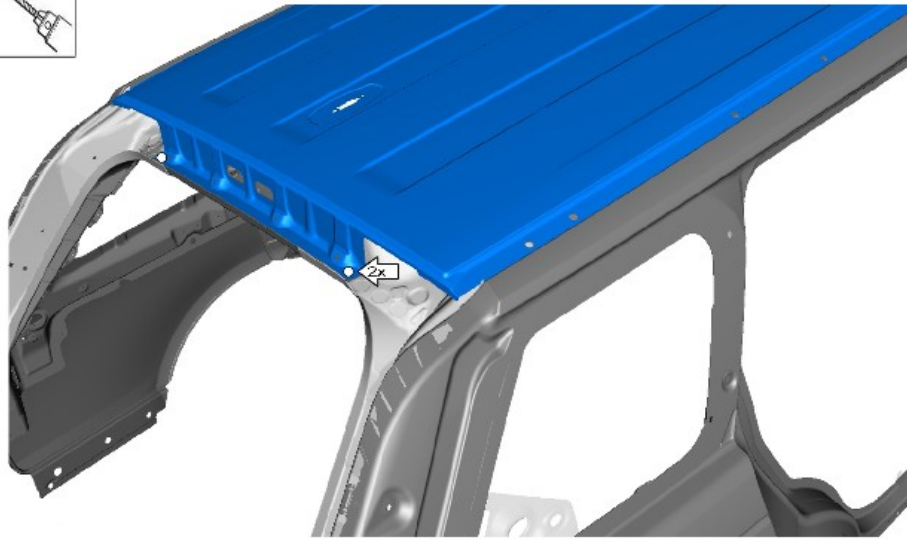
1. Remove rivet remnants.
2. Dress the flanges where necessary.
3. Using a fine bristle disc, clean and prepare the panel surfaces, removing any adhesive residue.
4. Offer up the new panel and clamp into position. Check alignment, if correct proceed to next step, if not, rectify and recheck before proceeding.
- 5.



E 143054

Drill holes through the old panel into the new panel at the points where BSF's are to be installed as indicated.

6.



E143055

Drill holes through the old panel into the new panel at the points where BSF's are to be installed as indicated.

7. Remove the new panel.

8. Deburr the drilled holes.

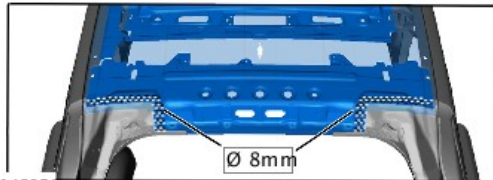
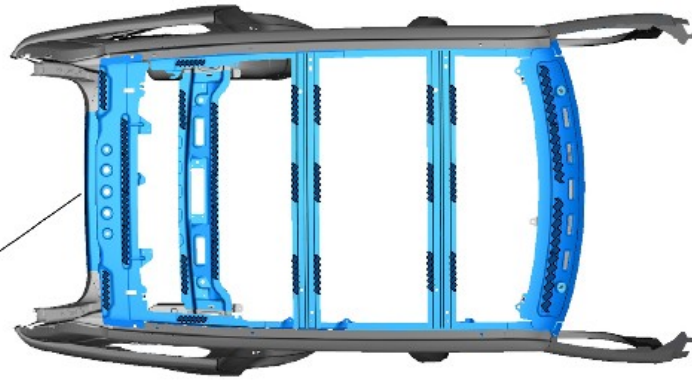
9. Using a fine bristle disc, clean and prepare the panel surfaces.

10. Apply the coupling agent where Land Rover recommended bonding material is to be applied and allow to dry.

11.

NOTE:

Make sure the areas highlighted have a sealer thickness of 8mm.



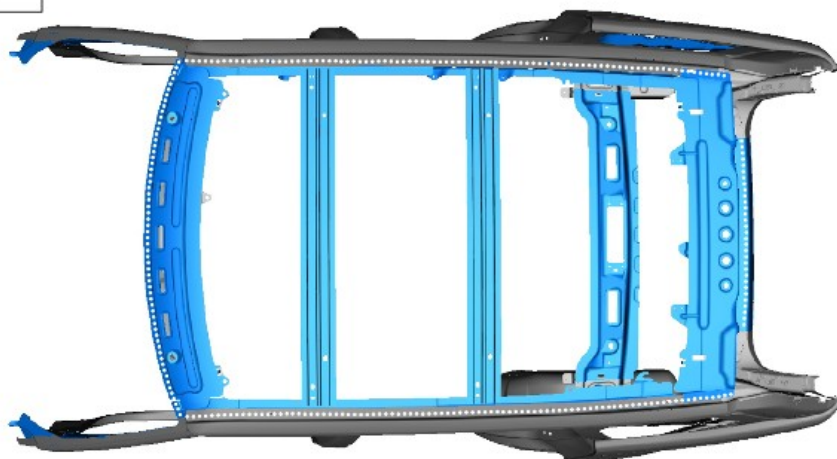
E143058

Apply semi-rigid sealer to the body at the points indicated.

12.

NOTE:

Make sure a continuous bead of adhesive surrounds fixing holes.



E143057

Apply a 5mm bead of Land Rover recommended bonding material to the body joints as indicated.

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13. Offer up the new panel, align and clamp into position. Offer up the new roof panel, clamp into position. Check alignment, if correct, proceed to next step, if not rectify and recheck before proceeding.
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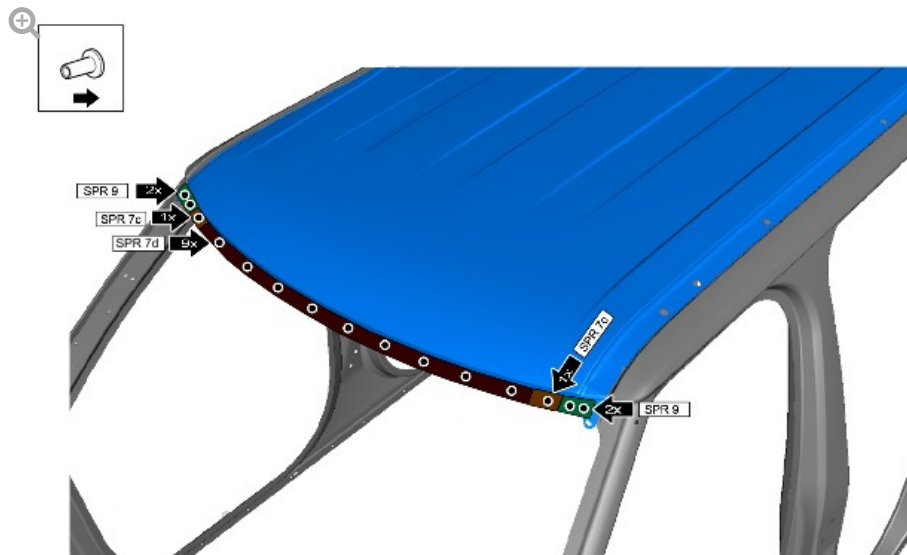
14.



E145212

Install the BSF's as indicated.

15.



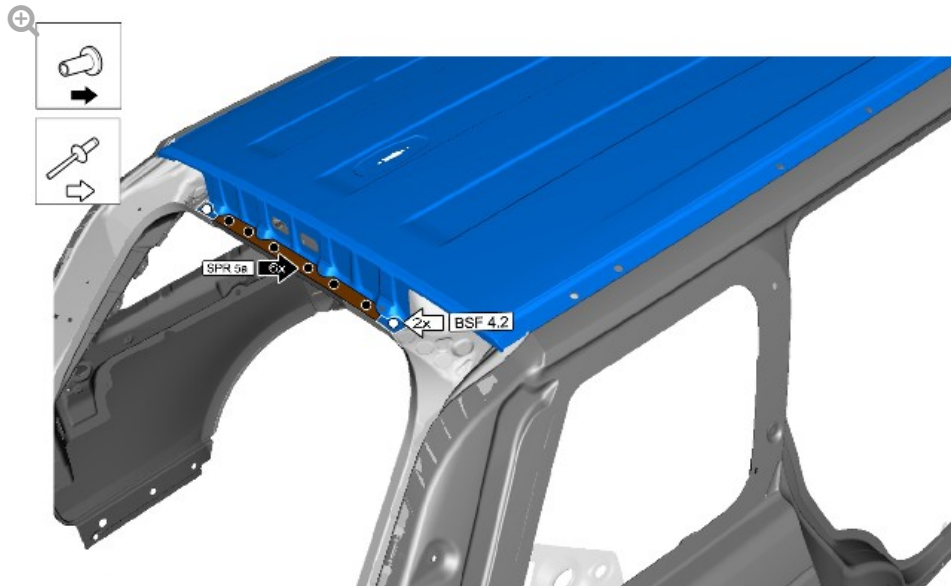
E143053

Install the SPR's as indicated.

16.

CAUTION:

Make sure that the BSF's are installed from inside outwards.



E 143052

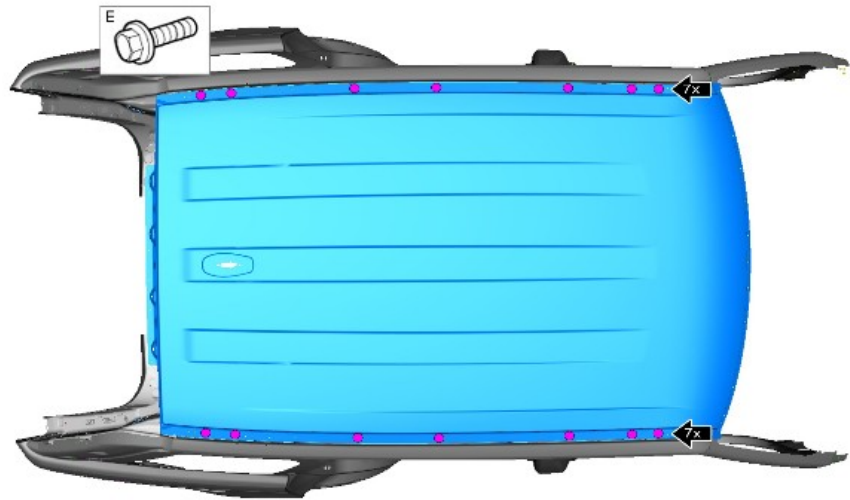
Install the SPR's and the BSF's as indicated.

17. Remove any excess adhesive.

18. Make sure that sealer is applied over the rivet heads including the remnants of the rivet stem and the surrounding hole.

19. Make sure that any open or exposed panel joints are suitably sealed.

20.



E144832

Install new bolts and tighten to 23Nm.

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21. Make sure corrosion protection is applied to all areas affected by the repair.
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22. The installation of associated panels and components is the reversal of removal procedure.